

4 (a) High temperature, Low pressure

- (b) Considering a single molecule colliding elastically against the container walls, which do not move upon collision, the molecule's impulse is equal to Δp , where p is the momentum of molecule. $\Delta p = mv - mu = -2mu$, where v is the velocity after collision $= -u$.

For constant velocity, the time interval, Δt , between collisions against a wall is $\Delta t = 2s/u$.

Using impulse-momentum theorem, the average force, F_{av} , that one molecule experiences due to the wall is given by

$$F_{av} = \frac{\Delta p}{\Delta t} = -\frac{mu^2}{s}$$

By N3L, the average force the molecule exerts on the wall

$$F_{\text{molecule on wall}} = -F_{av} = +\frac{mu^2}{s}$$

By definition of pressure, $P = \text{Force/Area}$. As the container has cross-sectional area A and side s , average pressure exerted by one molecule is given by

$$P = \frac{\left(\frac{mu^2}{s}\right)}{A} = \frac{mu^2}{V}$$

where the volume of container, $V = As$

(c)

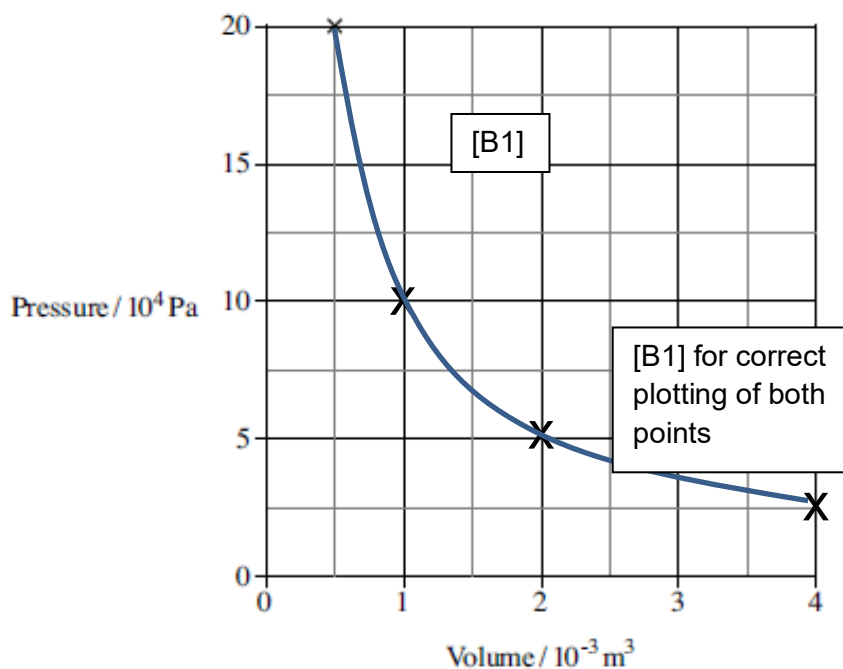


Fig. 4.2

- (ii) $pV = NkT \rightarrow N = (20 \times 10^4)(0.5 \times 10^{-3}) / (1.38 \times 10^{-23})(293.15)$
 $n = 0.3411$ moles
 $N = 2.47 \times 10^{22}$

- (iii) Microscopic KE $= 3/2 kT = 6.068 \times 10^{-21}$ J
 Total KE $= 3/2 NkT = 2.47 \times 10^{22} \times 6.068 \times 10^{-21}$
 $= 149.9$ J ~ 150 J